



INSTITUTE OF BUSINESS ADMINISTRATION

STATISTICAL INFERENCE ASSIGNMENT

Project Title: Statistical Analysis of Environmental Determinants of CO₂ Emissions per Capita

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Introduction

The growing concern over environmental degradation has made the analysis of carbon dioxide (CO₂) emissions a vital topic of global discussion because certain human activities such as industrialization, urbanization and extraction of resources have all led to the contribution of CO₂ emissions. The CO₂ emission per capita across countries are different due to various stages of economic development, industrial structure, population growth, and energy consumption ways. Identifying the driving factors behind CO₂ emissions would be imperative for the design of proper environmental and economic policies.

Therefore, in this project we will compare the level of CO₂ emissions in tons per year among countries that belong to four different income levels. The particular year that we used in our project is 2022.

This statistical investigation examines the relationship of CO₂ per capita emissions to selected economic and demographic variables for a set of countries. This project applies basic concepts of Introduction to Statistics (ITS) and Statistical Inference (SI) courses in investigating differences in CO₂ emissions among groups, and modeling the effect of selected predictors on CO₂ per capita emissions. In the analysis, we have included descriptive statistics, graphical summaries (histograms), hypothesis testing (t-tests taking the High-Income group as reference and the three hypothesis tests are: High-Income vs. Upper-Middle Income, High-Income vs. Lower-Middle Income, and High-Income vs. Low-Income and also ANOVA Analysis of variance to compare the means of the four income groups), and multiple regression modeling. Through these methods, we aim to identify significant determinants of CO₂ emissions and provide insights for sustainable policy formulation.

COMPARISON OF CO₂ EMISSIONS ACROSS DIFFERENT INCOME COUNTRY GROUPS

PART 1: DESCRIPTIVE/INFERENTIAL ANALYSIS

➤ DESCRIPTIVE STATISTICS: CO₂ Emissions among Income Groups

	<i>High Income</i>	<i>Upper Middle Income</i>	<i>Low Middle Income</i>	<i>Low Income</i>
Mean	194.0943	201.91935	24.0307	4.90535
Standard Error	49.52629726	50.23373623	5.839928339	1.297654683
Median	75.2455	114.4125	19.018	2.851
Standard Deviation	221.4883347	224.652098	26.1169535	5.803288163
Sample Variance	49057.08242	50468.56512	682.09526	33.6781535
Kurtosis	1.071537525	0.885698242	7.562922851	7.784322458
Skewness	1.464937854	1.127742934	2.663525227	2.492662135
Range	694.899	800.193	113.369	25.043
Minimum	28.432	0.554	0.494	0.271
Maximum	723.331	800.747	113.863	25.314
Sum	3881.886	4038.387	480.614	98.107
Count	20	20	20	20
Ratio of SD	1.14113776	1.112583306	1.086816177	1.183052823

The above table represents the descriptive statistics generated in Excel for four different income groups: High-Income, Upper-Middle Income, Lower-Middle Income, and Low-Income. In analyzing and comparing CO₂ emissions for the above groups, several statistical measures were considered, including the mean, median, skewness, and standard deviation.

The mean represents the average level of CO₂ emission in each income category. Based on the table, the average emission by the Upper-Middle Income group is the highest at 201.91 tons annually, and the High-Income group follows with an average of 194.09 tons per year. On the contrary, the Lower-Middle Income group records a significantly lower average emission at 24.03 tons, while the Low-Income class has the least mean value at 4.90 tons. This surely tells us that on average, higher-income countries emit substantially larger amounts of CO₂ compared to lower-income countries.

The median is the value that divides the dataset into two equal halves and supplies the central value of the data. In all these four categories, the Upper-Middle Income group has again recorded the highest median, 114.41, indicating that half of its countries emit below this value and half above it, while the Low-Income group has the lowest median of 2.851, which indicates that most of the countries in this category emit very little CO₂ compared to others.

As for the mode, there is no repeated value in any of the datasets from the different groups; therefore, there is no mode in any income group's CO₂ emission data.

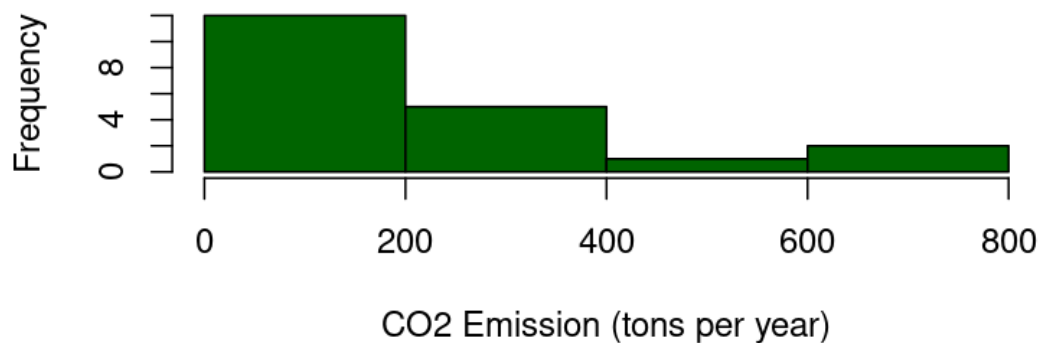
Skewness values describe the asymmetry or the direction of the tail of data. All groups show positive skewness. We noted the following values: 1.4649 for High-Income, 1.1277 for Upper-Middle Income, 2.6635 for Lower-Middle Income, and 2.4927 for Low-Income. This suggests that

in each group, most countries have lower CO₂ emissions while only a few countries have very high emissions, which pulls the tail of the distribution to the right.

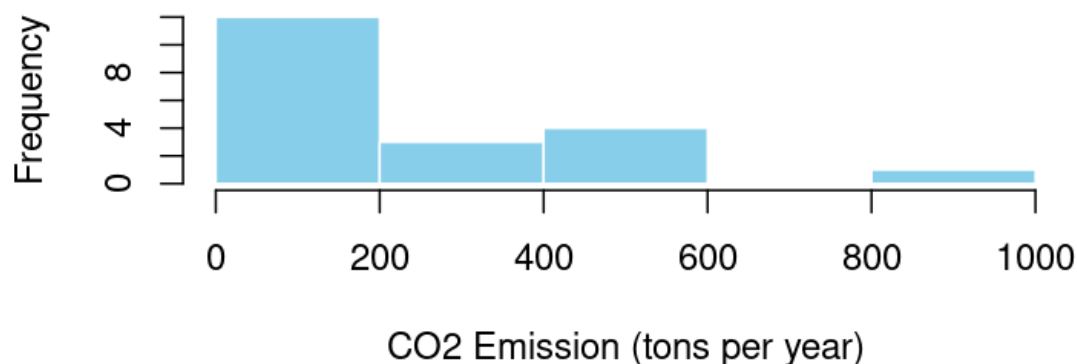
Finally, standard deviation describes the extent to which the values have varied from their means. Greater standard deviation shows larger variability among countries within a particular income group. According to the table, Upper-Middle Income has the highest variation with 224.65 in standard deviation and the High-Income group follows closely at 221.49. However, countries in the Lower-Middle Income category have a much lower spread of 26.11, and significantly lower, the Low-Income group, standing at 5.80, indicating that for low-income nations, CO₂ emissions are much more consistent and cluster closely around their mean values.

➤ HISTOGRAMS: CO₂ Emissions among Four Income Groups

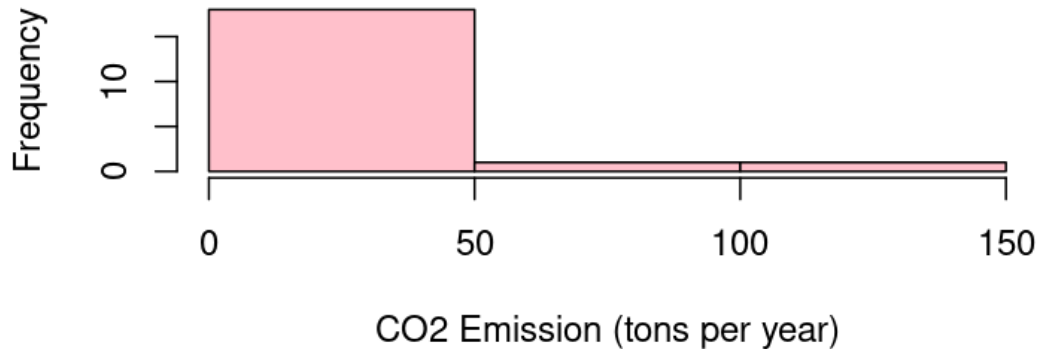
CO₂ Emission (High Income Countries)



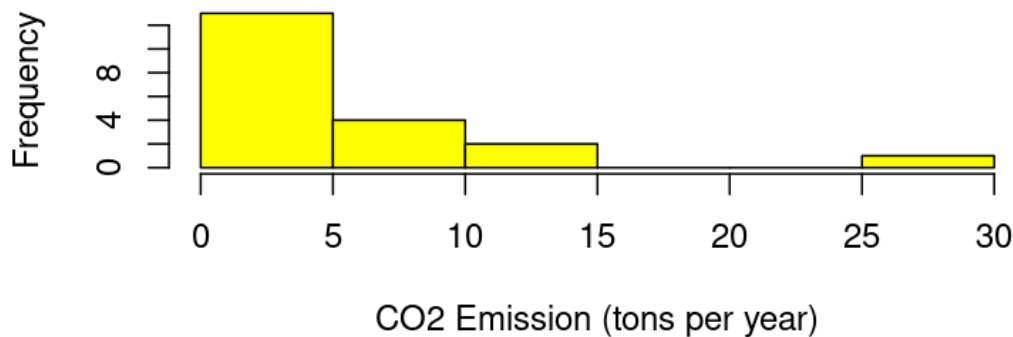
CO₂ Emission(Upper Middle Income Countries)



CO2 Emissions (Lower Middle Income Countries)



CO2 Emission (Low Income Countries)



Skewness in the histograms of all four income groups indicates that, for most countries, CO₂ emissions are lower, whereas for a minority of countries, higher values exist.

For the High-Income group, the histogram is positively skewed; the tail of the curve extends toward higher values of emissions. An outlier appears near the upper range, roughly between 400 and 600, which distorts the shape so that it is not perfectly skewed but leans to the right nonetheless.

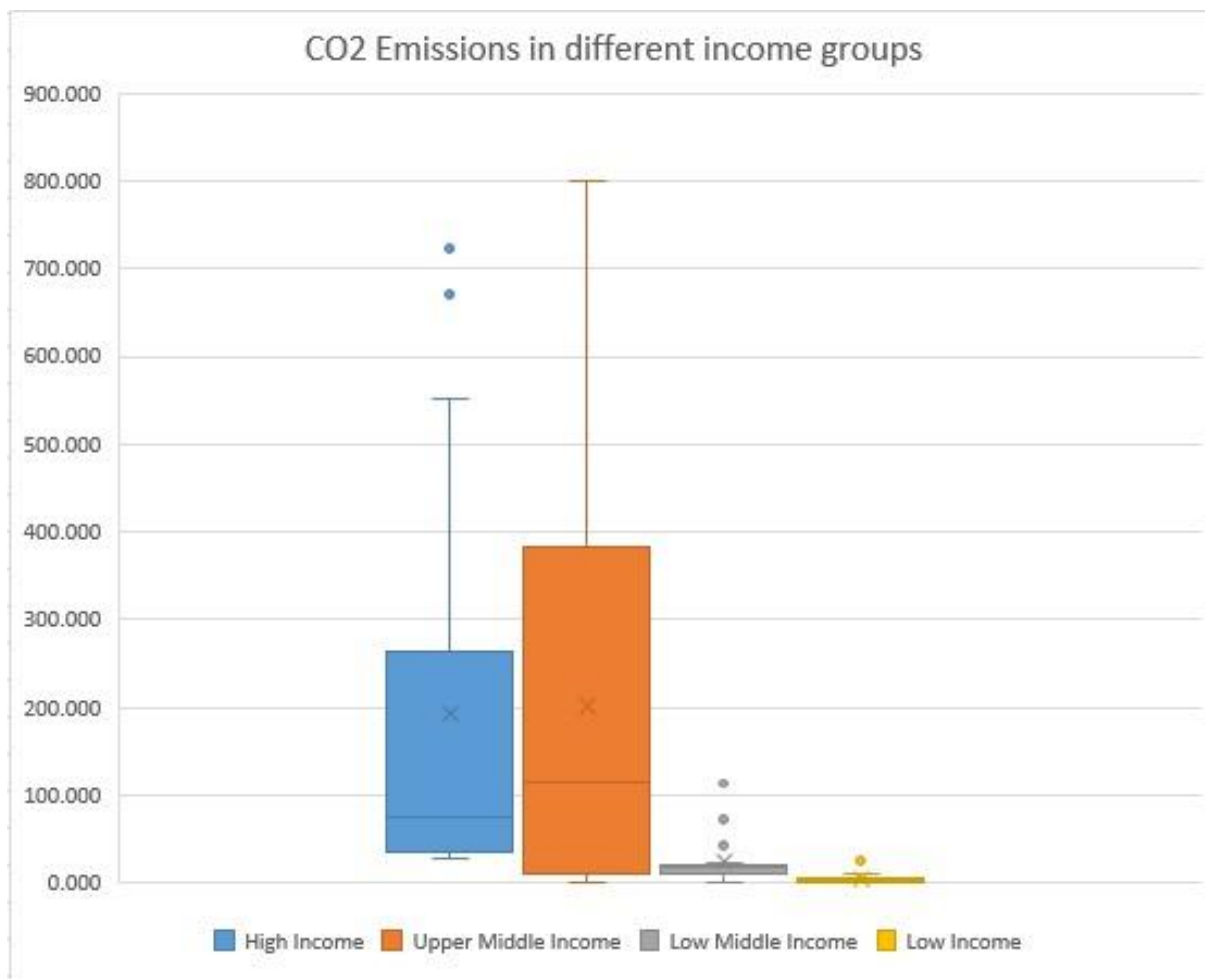
For the Upper-Middle Income category, the general pattern observed is the same. Its histogram is also right-skewed, with a clear outlier between 400 and 600. This one outlier represents the countries that have relatively higher CO₂ emissions than others.

Distribution for both Lower-Middle Income and Low-Income is also positively skewed, but their values of emission are far less. Within the Lower-Middle Income group of countries, there are two outliers between 50 and 150, which means that a couple of countries have a much higher emission

of CO₂ compared to others within the same income category. On the other hand, the Low-Income group contains only one clear outlier, coming somewhere in the area between 25 and 30, while the bulk of its countries stays toward the low end of the scale.

Altogether, all four histograms confirm that as income level decreases, the overall CO₂ emission level drops while the positive skew remains constant, showing that throughout all income groups, a small number of countries are responsible for relatively high emissions compared to the rest.

➤ BOX PLOT: CO₂ Emissions for Different Income Groups



A box and whisker plot describes the distribution of the data based on five particular values: minimum, first quartile, median, third quartile, and maximum, plus any outliers. The above box plot shows the CO₂ emissions for the four income groups. We have made them in different colors for easy visual identification: High-Income (in Blue Color), Upper-Middle Income (in Orange Color), Lower-Middle Income (in Gray Color), and Low-Income (in Yellow Color).

From the plot, it is evident that the largest range for CO₂ emissions belongs to the Upper-Middle Income category since the values in this category reach up to almost 800,000 tons. It also has the

highest median across all four categories, which implies that half of its countries have relatively large outputs of CO₂. The High-Income category follows with levels of emissions up to about 550,000 tons, but the median is somewhat lower compared with the Upper-Middle group. Much smaller levels of emissions were found in the Lower-Middle Income group as a whole, while the median was much closer to the bottom in this category. And finally, the Low-Income category results in the lowest emission, whose entire box lies close to the bottom of the chart, meaning very limited CO₂ production within these countries.

The “X” marks on each box represent the mean CO₂ emissions, and they confirm this pattern; the Upper-Middle group has the highest average, after which comes the High-Income group, while the Lower-Middle and Low-Income groups have much smaller averages.

Speaking of variation, the difference between the maximum and minimum values reflects the spacing of the emissions within each group. The Upper-Middle Income class shows the highest variability, which is supported by its tall whiskers and also wide box. The High-Income class has a little less variation but nevertheless covers a big range. The Lower-Middle Income class has a narrower range, showing less difference in emissions across its countries, while the Low-Income class has the least variation. Its box is compact, which means its countries are fairly similar in their levels of emissions.

In terms of shape, High-Income and Upper-Middle Income have positive skews: their top whisker is longer than their bottom one, and the mean lies closer to the bottom of each box. This indicates that most countries have lower emissions while few emit much more. The Lower-Middle Income group appears to have a slight positive skew as well, while it is less considerable. The range in the Low-Income group is too small to point out any obvious skew.

The plot also displays outliers, represented by small dots above the whiskers. There are two outliers for the High-Income group and one outlier for the Low-Income group. Three outliers are also fairly visible in the Lower-Middle Income category, showing countries that have high emissions for their respective classes. Rather surprisingly, not even a single outlier was seen within the large spread of the Upper-Middle Income category. Overall, the box plot confirms that CO₂ emissions increase with increase in income level and, even though richer countries contribute more in terms of emissions, their data also shows greater dispersion and outliers compared with lower-income nations, whose emissions are generally low and consistent.

➤ HYPOTHESIS TESTING

The purpose of the hypothesis testing is to see if the average CO₂ emissions within the High-Income group are different from that of the other three income groups. For this analysis, the High-Income group is considered as the reference category, and its mean emissions are contrasted with those of the Upper-Middle, Lower-Middle, and Low-Income groups.

High-Income vs Upper-Middle Income:

Let,

μ_1 = mean CO₂ emission of High-Income countries

μ_2 = mean CO₂ emission of Upper-Middle Income countries

(1) $H_0: \mu_1 = \mu_2$

$H_a: \mu_1 \neq \mu_2$

(2) Non-pooled t test

In order to compare the two means, we've completed a two-sample t-test in Excel assuming unequal variances (non-pooled test). This was appropriate since the larger standard deviation divided by the smaller one was greater than 1.5, indicating that the two groups are not equally variable. The test was conducted at a confidence level of 95%, and therefore $\alpha = 0.05$.

t-Test: Two-Sample Assuming Unequal Variances		
	<i>High Income</i>	<i>Upper Middle Income</i>
Mean	194.0943	201.91935
Variance	49057.08242	50468.56512
Observations	20	20
Hypothesized Mean Difference	0	
df	38	
t Stat	-0.110926322	
P(T<=t) one-tail	0.456129298	
t Critical one-tail	1.68595446	
P(T<=t) two-tail	0.912258595	
t Critical two-tail	2.024394164	

(3) From the output table, the t-statistic is -0.1109, and the critical t-value (two-tailed) is ± 2.024 . Since the calculated t-statistic lies within the acceptance region, between -2.024 and +2.024, the null hypothesis cannot be rejected. Additionally, the P-value of 0.9122 (two-tailed) is significantly greater than the alpha level of 0.05, supporting the fact that there is no statistical difference between the two groups of means.

(4) Do not Reject H_0

At 5% level of significance, the data does not provides sufficient evidence to conclude that the mean CO₂ Emissions of high-income countries differ from upper-middle income countries. This shows that there is no significant difference between the average CO₂ emissions in High-Income and Upper-Middle Income countries. That is, the two groups have approximately equal mean emissions.

High-Income vs Lower-Middle Income:

Let,

μ_1 = mean CO₂ emission of High-Income countries

μ_2 = mean CO₂ emission of Lower-Middle Income countries

(1) $H_0: \mu_1 = \mu_2$

$H_a: \mu_1 \neq \mu_2$

(2) Non-pooled t test

In order to compare the two means, we've completed a two-sample t-test in Excel assuming unequal variances (non-pooled test). This was appropriate since the larger standard deviation divided by the smaller one was greater than 1.5, indicating that the two groups are not equally variable. The test was conducted at a confidence level of 95%, and therefore $\alpha = 0.05$.

t-Test: Two-Sample Assuming Unequal Variances		
	<i>High Income</i>	<i>Lower Middle Income</i>
Mean	194.0943	24.0307
Variance	49057.08242	682.09526
Observations	20	20
Hypothesized Mean Difference	0	
df	20	
t Stat	3.410178136	
P(T<=t) one-tail	0.001387715	
t Critical one-tail	1.724718243	
P(T<=t) two-tail	0.00277543	
t Critical two-tail	2.085963447	

(3) From the output table, the t-statistic is 3.410, and the critical t-value (two-tailed) is ± 2.0859 . Since the calculated t-statistic falls in the rejection region, the null hypothesis is rejected. Additionally, the P-value of 0.0013 (two-tailed) is significantly smaller than the alpha level of 0.05, supporting the fact that there is statistical difference between the two groups of means.

(4) Reject H_0

At 5% level of significance, the data provides sufficient evidence to conclude that the mean CO₂ Emissions of high-income countries differ from lower-middle income countries. This shows that there is significant difference between the average CO₂ emissions in High-Income and Lower-Middle Income countries.

High-Income vs Low-Income:

Let,

μ_1 = mean CO₂ emission of High-Income countries

μ_2 = mean CO₂ emission of Low Income countries

(1) $H_0: \mu_1 = \mu_2$

$H_a: \mu_1 \neq \mu_2$

(2) Non-pooled t test

In order to compare the two means, we've completed a two-sample t-test in Excel assuming unequal variances (non-pooled test). This was appropriate since the larger standard deviation divided by the smaller one was greater than 1.5, indicating that the two groups are not equally variable. The test was conducted at a confidence level of 95%, and therefore $\alpha = 0.05$.

t-Test: Two-Sample Assuming Unequal Variances		
	<i>High Income</i>	<i>Low Income</i>
Mean	194.0943	4.90535
Variance	49057.08242	33.6781535
Observations	20	20
Hypothesized Mean Difference	0	
df	19	
t Stat	3.818659053	
P(T<=t) one-tail	0.000579664	
t Critical one-tail	1.729132812	
P(T<=t) two-tail	0.001159328	
t Critical two-tail	2.093024054	

(3) From the output table, the t-statistic is 3.8186, and the critical t-value (two-tailed) is ± 2.093 . Since the calculated t-statistic falls in the rejection region, the null hypothesis is rejected. Additionally, the P-value of 0.005 (two-tailed) is significantly smaller than the alpha level of 0.05, supporting the fact that there is statistical difference between the two groups of means.

(4) Reject H_0

At 5% level of significance, the data provides sufficient evidence to conclude that the mean CO₂ Emissions of high-income countries differ from low-income countries. This shows that there is significant difference between the average CO₂ emissions in High-Income and Low-Income countries.

➤ ANOVA TEST

The ANOVA test is used to identify whether there is a significant difference between means across more than two groups. In this project, the test will be applied to see if the mean CO₂ emissions in the four income categories are significantly different or relatively similar. The following outlines how to go about conducting the test, along with the Excel output:

Anova: Single Factor							
SUMMARY							
Groups	Count	Sum	Average	Variance			
High Income	20	3881.886	194.0943	49057.08242			
Upper Middle Income	20	4038.387	201.91935	50468.56512			
Low Middle Income	20	480.614	24.0307	682.09526			
Low Income	20	98.107	4.90535	33.6781535			
ANOVA							
Source of Variation	SS	df	MS	F	P-value	F crit	
Between Groups	677999.9263	3	225999.9754	9.018227128	3.51082E-05	2.72494392	
Within Groups	1904586.998	76	25060.35524				
Total	2582586.924	79					

Hypothesis Test and Analysis:

Let,

μ_1 = mean CO₂ emission for High-Income countries

μ_2 = mean CO₂ emission for Upper-Middle income countries

μ_3 = mean CO₂ emission for Lower-Middle income countries

μ_4 = mean CO₂ emission for Low-Income countries

(1) $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ (All four income countries have the same average CO₂ emission)

$H_a: \mu_1 = \mu_2$ (At least two countries have the same mean)

(2) $F = 9.018$

(3) Critical Region = 2.725

The calculated F-value falls within the acceptance region only when it is smaller than the critical F-value. However, if the calculated F-value exceeds the critical value, it falls in the rejection region. In the output provided, the F-calculated is 9.018, whereas the critical F-value is 2.725. Since 9.018 is greater than 2.725, we must reject the null hypothesis. Additionally, the P-value (two-tailed) is significantly smaller than the alpha level of 0.05, supporting this fact.

(4) Reject H_0

At 5% significance level, the data provides sufficient evidence to conclude that there is a significant difference between the mean CO₂ emissions (in tons per year) among the four income countries.

PART 2: REGRESSION ANALYSIS

MULTIPLE REGRESSION: CO₂ Emissions among Income Groups

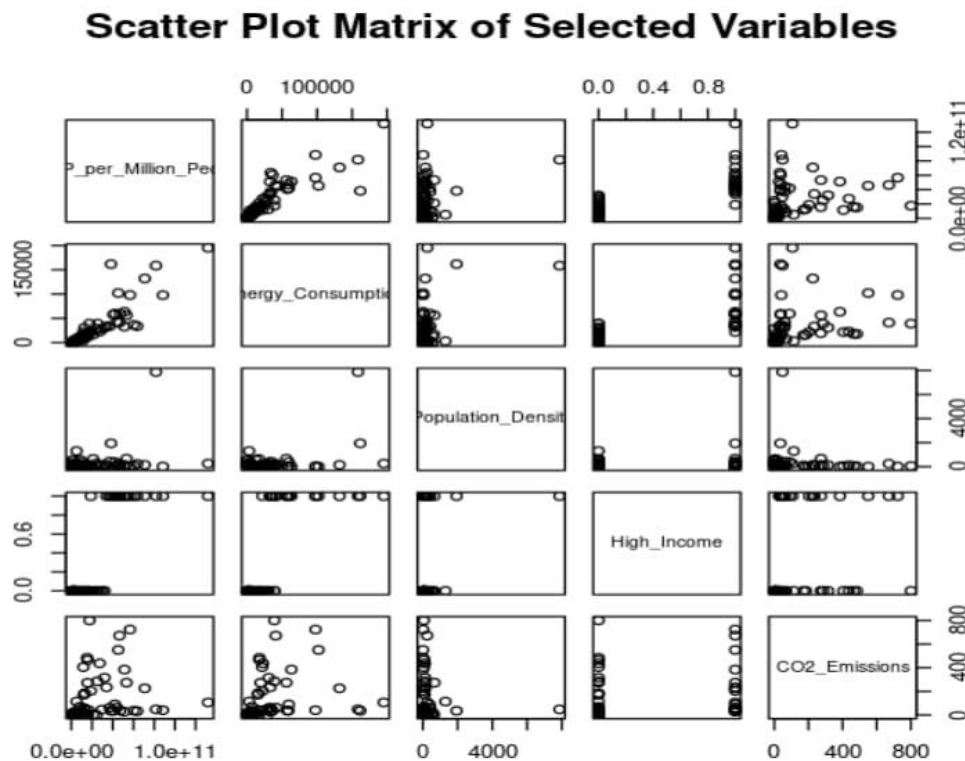
In multiple regression, two or more independent variables are used to explain one dependent variable. It is a statistical technique that examines how one dependent variable is influenced by two or more independent variables.

This study applies multiple regression in an attempt to analyze the determinants of CO₂ emissions, taken as the dependent variable. Other independent variables involved in the model are energy consumption per capita, population density, and a dummy variable for income level. The dummy takes the value 1 for high-income countries and 0 for all other income groups, enabling the model to capture structural differences between high-income and non-high-income countries.

➤ SCATTER PLOT MATRIX

The scatter plot matrix provides an initial view of how the dependent variable of CO₂ emissions relate to the selected predictors which are the independent variables. These are shown in summary form in the following scatter plot matrix. There is a noticeable positive relationship between CO₂ emissions and energy consumption, CO₂ emissions go up when energy use is higher. In contrast, population density shows almost no clear pattern, as most countries cluster at low density levels

with little change in emissions. The relationship between GDP per million people and emissions appears to be very weak, with points scattered randomly and no visible trend. Lastly, CO₂ emissions show greater variability among high-income countries, that high-income countries exhibit more variation in CO₂ emissions level compared to the other income groups.



➤ STANDARD OUTPUT

The corresponding Excel regression output is presented below:

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.395903203					
R Square	0.156739346					
Adjusted R Square	0.111765445					
Standard Error	170.4104752					
Observations	80					
ANOVA						
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>	
Regression	4	404827.5247	101206.8812	3.485117836	0.011508367	
Residual	75	2177979.755	29039.73006			
Total	79	2582807.279				
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	68.49710944	24.80220894	2.761734232	0.007225989	19.0885756	117.9056433
GDP(US dollars Per million People)	-1.96481E-10	1.98781E-09	-0.098843203	0.921526475	-4.1564E-09	3.76343E-09
Energy Consumption (kWh per person per year)	0.001934988	0.001128321	1.714926741	0.090488006	-0.000312743	0.004182719
Population Density(People per km ²)	-0.050409526	0.024385915	-2.067157489	0.042170234	-0.098988759	-0.001830293
Dummy Variable(High-Income)	21.56066809	78.98757227	0.27296279	0.785632016	-135.7906448	178.9119809

➤ ESTIMATED EQUATION

$$\text{CO}_2 \text{ Emissions} = 68.4971 - 1.9648 \times 10^{-10} (\text{GDP}) + 0.0019(\text{Energy Consumption}) - 0.0504(\text{Population Density}) + 21.5607(\text{High-Income})$$

- CO₂ Emissions: tons per year
- GDP: US dollars per million people
- Energy Consumption: kWh per person per year
- Population Density: people per km²
- High-Income: dummy variable (1 = high-income group, 0 = not)

➤ INTERPRETATION OF COEFFICIENTS

Intercept (68.4971): A country with 0 GDP, 0 energy consumption, 0 population density, and does not belongs to the high-income group has a CO₂ emission approximately equal to 68.4971 tons per year. Although it's unrealistic but serves as baseline level of emissions. It has no real-world interpretation because such conditions are impossible.

Coefficient of GDP per capita (-1.9648×10^{-10}): Keeping energy consumption, population density, and income group fixed, if GDP increases by 1 US dollar per million people, the CO₂ emission of a country will decrease by 1.9648E-10 tons per year.

Coefficient of Energy consumption per capita (0.0019): Keeping GDP, population density, and income group fixed, if the energy consumption of a country increases by 1 kilowatt-hour per person per year, the CO₂ emission will increase by 0.0019 tons per year.

Coefficient of Population density (0.0504): Keeping GDP, energy consumption, and income group fixed, if the population density of a country increases by 1 person per km², its CO₂ emission will decrease by 0.0504 tons per year.

Coefficient of High-Income group (21.5607): Keeping GDP, energy consumption, and population density fixed, if a country belongs to High-Income group, its CO₂ emission will be 21.5607 tons per year higher than a country belonging to any other income group.

➤ INTERPRETATION OF R²

Coefficient of Determination (0.1567): Approximately, 15.67% of the variations in the CO₂ emissions (tons per year) across countries is explained by the independent variables GDP, energy consumption per capita, population density and income group.

➤ HYPOTHESIS TESTING

GDP per Capita

Let,

(1) $H_0: \beta_1 = 0$ (GDP per Capita is not a useful predictor of CO₂ emissions)

$H_a: \beta_1 \neq 0$ (GDP per Capita is a useful predictor of CO₂ emissions)

(2) $t = -0.0988$ (from excel output)

(3) $DF = n - (k+1)$

$$= 80 - (4 + 1) = 75$$

$$\alpha = 0.05/2 = 0.025$$

$$t(0.025, 75) = 1.992$$

(4) Do not reject H_0

At 5% significance level, the data does not provides sufficient evidence to conclude that the GDP (US dollars per million people) is a useful predictor of CO₂ emissions (tons per year). Additionally, the P-value of 0.9215 (two-tailed) is significantly greater than the alpha level of 0.05, supporting the fact that GDP per Capita is not a useful predictor of CO₂ emissions.

Energy Consumption per Capita

Let,

(1) $H_0: \beta_2 = 0$ (Energy Consumption per Capita is not a useful predictor of CO₂ emissions)

$H_a: \beta_2 \neq 0$ (Energy Consumption per Capita is a useful predictor of CO₂ emissions)

(2) $t = 1.7149$ (from excel output)

(3) $DF = n - (k+1)$

$$= 80 - (4 + 1) = 75$$

$$\alpha = 0.05/2 = 0.025$$

$$t(0.025, 75) = 1.992$$

(4) Do not reject H_0

At 5% significance level, the data does not provides sufficient evidence to conclude that the Energy Consumption (in kilowatt-hours (kWh) per person per year) is a useful predictor of CO₂ emissions (tons per year). Additionally, the P-value of 0.0904 (two-tailed) is significantly greater than the alpha level of 0.05, supporting the fact that Energy Consumption per Capita is not a useful predictor of CO₂ emissions.

Population Density

Let,

(1) $H_0: \beta_3 = 0$ (Population density is not a useful predictor of CO₂ emissions)

$H_a: \beta_3 \neq 0$ (Population Density is a useful predictor of CO₂ emissions)

(2) $t = -2.0671$ (from excel output)

(3) $DF = n - (k+1)$

$$= 80 - (4 + 1) = 75$$

$$\alpha = 0.05/2 = 0.025$$

$$t(0.025, 75) = 1.992$$

(4) Reject H_0

At 5% significance level, the data provides sufficient evidence to conclude that the Population Density (people per km²) is a useful predictor of CO₂ emissions (tons per year). Additionally, the P-value of 0.0423 (two-tailed) is significantly smaller than the alpha level of 0.05, supporting the fact that Population Density is a useful predictor of CO₂ emissions.

Income Group

Let,

(1) $H_0: \beta_4 = 0$ (Income Group is not a useful predictor of CO₂ emissions)

$H_a: \beta_4 \neq 0$ (Income Group is a useful predictor of CO₂ emissions)

(2) $t = 0.2730$ (from excel output)

(3) $DF = n - (k+1)$

$$= 80 - (4 + 1) = 75$$

$$\alpha = 0.05/2 = 0.025$$

$$t(0.025, 75) = 1.992$$

(4) Do not Reject H_0

At 5% significance level, the data does not provides sufficient evidence to conclude that the Income Group (High-Income or not) is a useful predictor of CO₂ emissions (tons per year). In addition, the p-value is 0.7856 which is greater than 0.05. Additionally, the P-value of 0.7856 (two-tailed) is significantly greater than the alpha level of 0.05, supporting the fact that Income Group is not a useful predictor of CO₂ emissions.

➤ PREDICTIONS AND RESIDUAL TABLE

We took the 3rd country from each Income group (High, Upper Middle, Lower Middle and Low) to find the estimated CO₂ emissions using their GDP per capita, energy consumption per capita and population density for the year 2022. After that we have compared the predicted values with the actual values of CO₂ emissions for those countries to make predictions.

1 = Austria (High Income)

- GDP (US dollars per million people) = 43,996,637,480.49
- Energy consumption (kWh per person per year) = 42122.105
- Population density (people per km²) = 110
- Dummy variable (High-Income) = 1

Estimated CO₂ Emissions = $68.4971 - 1.9648 \times 10^{-10} (\text{GDP}) + 0.0019(\text{Energy Consumption}) - 0.0504(\text{Population Density}) + 21.5607(\text{High-Income})$

= $68.4971 - 1.9648 \times 10^{-10} (43,996,637,480.49) + 0.0019(42122.105) - 0.0504(110) + 21.5607(1)$
= 155.8858 tons per year

Actual CO₂ Emissions = 61.489 tons per year

Hence, the regression model has overestimated the CO₂ Emissions.

2 = Botswana (Upper Middle Income)

- GDP (US dollars per million people) = 16,104,778,436.78
- Energy consumption (kWh per person per year) = 12002.065
- Population density (people per km²) = 4
- Dummy variable (High-Income) = 0

Estimated CO₂ Emissions = $68.4971 - 1.9648 \times 10^{-10} (\text{GDP}) + 0.0019(\text{Energy Consumption}) - 0.0504(\text{Population Density}) + 21.5607(\text{High-Income})$

= $68.4971 - 1.9648 \times 10^{-10} (16,104,778,436.78) + 0.0019(12002.065) - 0.0504(4) + 21.5607(0)$
= 87.9350 tons per year

Actual CO₂ Emissions = 6.596 tons per year

Hence, the regression model has overestimated the CO₂ Emissions.

3 = Bolivia (Lower Middle Income)

- GDP (US dollars per million people) = 6,476,013,885.19
- Energy consumption (kWh per person per year) = 8554.576
- Population density (people per km²) = 11
- Dummy variable (High-Income) = 0

Estimated CO₂ Emissions = $68.4971 - 1.9648 \times 10^{-10} (\text{GDP}) + 0.0019(\text{Energy Consumption}) - 0.0504(\text{Population Density}) + 21.5607(\text{High-Income})$

= $68.4971 - 1.9648 \times 10^{-10} (6,476,013,885.19) + 0.0019(8554.576) - 0.0504(11) + 21.5607(0)$

= 82.9241 tons per year

Actual CO₂ Emissions = 22.378 tons per year

Hence, the regression model has overestimated the CO₂ Emissions.

4 = Burundi (Low Income)

- GDP (US dollars per capita) = 689,164,790.73
- Energy consumption per million people = 273

- Population density = 519
- Dummy variable = 0

Estimated CO₂ Emissions = $68.4971 - 1.9648 \times 10^{-10} (\text{GDP}) + 0.0019(\text{Energy Consumption}) - 0.0504(\text{Population Density}) + 21.5607(\text{High-Income})$

= $68.4971 - 1.9648 \times 10^{-10} (689,164,790.73) + 0.0019(273) - 0.0504(519) + 21.5607(0)$

= 42.7228 tons per year

Actual CO₂ Emissions = 0.707 tons per year

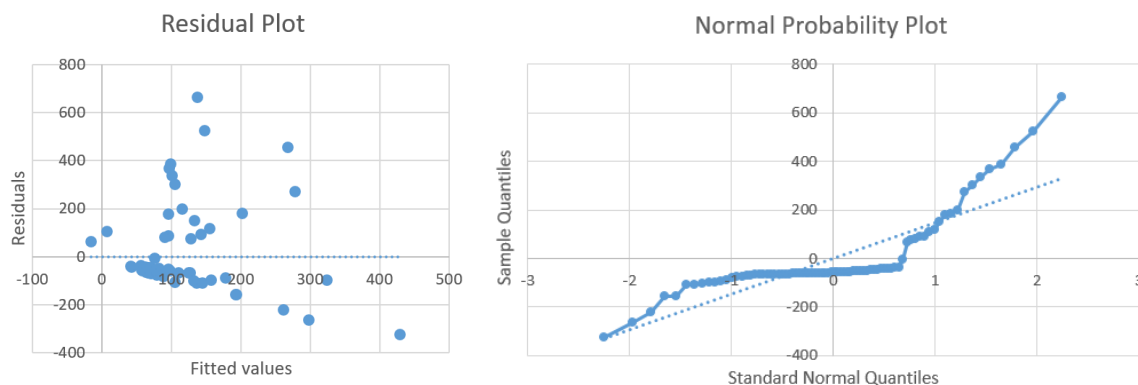
Hence, the regression model has overestimated the CO₂ Emissions.

Prediction table:

Country	Actual CO ₂ Emissions (tons per year)	Estimated CO ₂ Emissions (tons per year)	Residual = Actual - Predicted
Austria	61.489	155.8858	-94.3968
Botswana	6.596	87.9350	-81.339
Bolivia	22.378	82.9241	-60.5461
Burundi	0.707	42.7228	-42.0158

These predictions show collectively that the model has overestimated the values of CO₂ Emissions. This means that it is not perfect because there are other important variables that are not used in the model but can explain the bulk of the CO₂ Emissions' variation from country to country.

➤ RESIDUAL AND NORMAL PROBABILITY PLOTS



- The residual plot indicates that the linearity assumption is upheld because the residuals are randomly scattered with no obvious pattern. However, the residuals show a widening spread as the fitted values increase, which means the constant variance assumption is violated. This suggests that the reliability of the model's estimates (standard errors, p-values) may be compromised.
- The normal probability plot shows that the points deviate significantly from the dotted straight line, indicating that the normality assumption is violated. The distribution primarily shows heavy tails along with the long tails stretching far from the line in the

extremes. In addition, the presence of potential outliers further shows the violation of normality assumption.

Conclusion

From the descriptive analysis, it can be concluded that the High and Upper Middle income countries have the highest average CO₂ emissions, at 194.0943 and 201.9193 respectively, while the Lower Middle and Low income countries have much lower averages, at 24.0307 and 4.9053. High and Upper Middle income groups also show the largest variation in emissions with a standard deviation of 221.4883 and 224.6521 respectively, whereas Low income countries have the most consistent levels having a standard deviation of 5.8033. High income countries have a slight positive skew (1.4649), with most emitting less CO₂ and a few much more. Upper Middle income countries show a skewness of 1.1277, while Lower Middle and Low income countries are more skewed (2.6635 and 2.4927), meaning most emit very low amounts. Boxplots confirm this: High and Upper Middle income countries have wide spreads, with Upper Middle income ones show the most variation. Low income countries have very low emissions with minimal variation and only one outlier. Hypothesis testing indicates significant differences in mean emissions between High income countries and the other three groups.

However, the regression analysis shows that the model explains just 15.67% of the variation, suggesting that there are other factors that also play an important role. We saw that at 5% of significance level, only Population Density is a significant variable which is greatly related to CO₂ emissions. This shows that the regression model is a poor predictor of CO₂ emissions. Thus, we should incorporate variables such as industrialization and policies into the data for better results.

Analysis in R

Histograms:

```
> hist(
+   high_income$`CO2 Emission tons per year`,
+   main = "CO2 Emission (High Income Countries)",
+   xlab = "CO2 Emission (tons per year)",
+   col = "darkgreen",
+   border = "black",
+   breaks = 4
+ )

> h <- hist(
+   upper_middle$`CO2 Emission tons per year`,
+   main = "CO2 Emission(Upper Middle Income Countries)",
+   xlab = "CO2 Emission (tons per year)",
+   col = "skyblue",
+   border = "white",
+   breaks = 4
+ )
> |

> h <- hist(
+   lower_middle$`CO2 Emission tons per year`,
+   main = "CO2 Emissions (Lower Middle Income Countries)",
+   xlab = "CO2 Emission (tons per year)",
+   col = "pink",
+   border = "black",
+   breaks = 4
+ )
> |

> h <- hist(
+   low_income$`CO2 Emission tons per year`,
+   main = "CO2 Emission (Low Income Countries)",
+   xlab = "CO2 Emission (tons per year)",
+   col = "yellow",
+   border = "black",
+   breaks = 4
+ )
> |
```

Scatter Plot Matrix:

```
Session restored from your saved work on 2025-Dec-09 08:34:05 UTC (7 hours ago)
> pairs(
+   ~ GDP_per_Million_People + Energy_Consumption + Population_Density + High_In
come + CO2_Emissions,
+   data = SI_clean,
+   main = "Scatter Plot Matrix of Selected Variables"
+ )
> |
```

References

- All the data for the year 2022 on CO₂ Emissions has been taken from the following links:

<https://github.com/owid/co2-data>

https://en.wikipedia.org/wiki/List_of_countries_by_energy_consumption_per_capita

https://en.wikipedia.org/wiki/List_of_countries_and_dependencies_by_population_density

- Descriptive Statistics, Box plot, Hypothesis Testing, ANOVA Test, Summary for Multiple Regression, Residual Plot and Normal Probability Plot are done on Microsoft Excel 2013.
- Histograms and Scatter Plot Matrix have been made on R Studio.